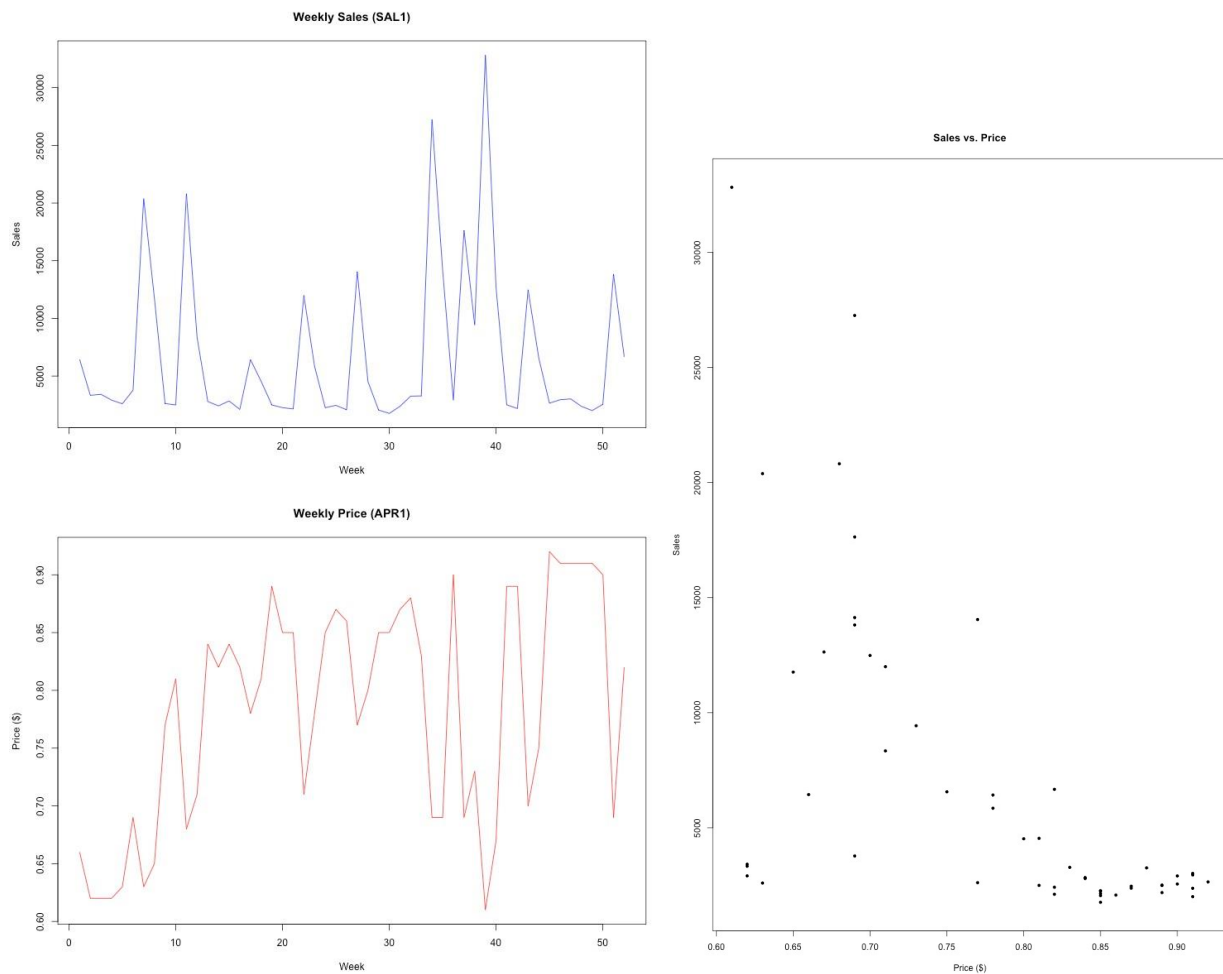


Problem 3.31

(a)

Summary statistics for SAL1 and APR1 :

Variable	Mean	Std	Min	Median	Max
SAL1	6,719	6,915.08	1,762	3,136	32,820
APR1	0.7825	0.09804	0.61	0.81	0.92



(b)

Higher prices are associated with lower unit sales. This is consistent with the law of demand — as price increases, quantity demanded falls. The negative relationship is expected from an economics standpoint.

(c)

Define PRICE1 = 100 · APR1 (price in cents). The estimated regression is:

$$\text{SAL1} = 40,714.22 - 434.45 \cdot \text{PRICE1}$$

Point estimate: $b_2 = -434.45$. A one-cent increase in the price of brand no. 1 is estimated to decrease sales by 434.45 units on average.

95% Confidence Interval for β_2 :

$$b_2 \pm t_{0.025,50} \times \text{se}(b_2) = -434.45 \pm 2.009 \times 78.58$$

$$95\% \text{ CI: } (-592.29, -276.60) \text{ [units per cent]}$$

(d)

When price per can = 70 cents, PRICE1 = 70:

$$\text{SAL1} = 40,714.22 - 434.45 \times 70 = 40,714.22 - 30,411.5 = 10,302.9 \text{ units}$$

$$90\% \text{ CI for } E[\text{SAL1} \mid \text{PRICE1} = 70]: (8,624.9, 11,980.9) \text{ [units]}$$

(e)

Point elasticity of SAL1 with respect to PRICE1 at the means:

$$\hat{\epsilon} = b_2 \times (\text{APRICE1} / \text{ASAL1}) = -434.45 \times (78.25 / 6,719) = -5.060$$

$$95\% \text{ CI for elasticity: } (-6.898, -3.222)$$

Since $|\hat{\epsilon}| = 5.060 > 1$, sales are fairly elastic with respect to price. This makes economic sense: canned tuna is a commodity with readily available substitutes, so consumers are sensitive to price changes.

(f)

Hypotheses (in terms of the elasticity $\epsilon = \beta_2 \times (\text{APRICE1} / \text{ASAL1})$):

$$H_0: \epsilon = -3$$

$$H_1: \epsilon \neq -3$$

Test statistic:

$$t = (\hat{\epsilon} - (-3)) / \text{se}(\hat{\epsilon}) = (-5.060 - (-3)) / \text{se}(\hat{\epsilon}) = -2.251$$

$$\text{Rejection region: } |t| > t_{0.05,50} = 1.676$$

$$\text{p-value: } 0.0288$$

Since $|-2.251| = 2.251 > 1.676$ and $\text{p-value} = 0.0288 < 0.10$, we reject H_0 at the 10% level of significance.

The data do not support the claim that the elasticity equals -3 . The estimated elasticity (-5.060) is significantly different from -3 .